and validation
at which
* It learns our interests.
* It improves future recommendations

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Assignment-1

1. Components of a learnings problem

(i) Representation:

It means deciding how the m/c model will repeat the problem and learn from the data

The model uses

- Features (I/P) - Information given to the model
- Target (O/P) - What we want to predict

Eg:- House price prediction

The model learns the relationship b/w the i/p and o/p

Ex:- Decision tree Linear regression. Neural
- Support vector machine

(ii) Evaluation

- Checking how well m/c model performs.
- We compare the models predictions with the actual answers

Eg:- Actual price = 50L
predicted price = 48L

(iii) Optimization:
Finding the best model pa...
better predictions

The model

1. make prediction
2. calculates the error
3. Adjust the parameters
4. makes better prediction

The process is repeted until the error is reduced.

2. perspectives and issues in ML

1. Learning from data
2. Learning from experience
3. statistical perpective
4. Computation perspective

1. Learning From data

ML systems Learns pattern from data instead of being explicity programmed for every situation

Eg:- If we provide many emails label

- Spam
- Not Spam

Assignment-1

-as problem

· Learning from experience:

+ ML · System can improve its performance through experience

Eg:- Recommendation system

- The system observes our activity
- It learns our interests

Statistical perspective:

ML uses :statistical and prob to find pattern in data for example: a model ·calculate prob

A student will pass based on attendance

+ final relationships

+ measure uncertainity

+ compare models

Computation perspective:

ML is also a computation problem

The computer must

+ process ·large amts of data

+ Select a suitable algorithm

+ Train the model

+ perform prediction efficiently